Prakhar Singh

Bachelor of Technology
Electronics & Communication Engineering
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Education

Degree	Institute	CGPA	Year
B.Tech., ECE	Birla Institute of Technology,Mesra Patna	6.50	2022-2026

Experience

- **Central Coalfields Limited** *July 2025 - Aug 2025*
  Software Development Intern (CampusConnect) Ranchi,Jharkhand, India
  - Developed and deployed a full-stack event management platform (CampusConnect) using Spring Boot, Hibernate, and React.js, supporting 50+ active users.
  - Designed and implemented RESTful APIs for event registration, user management, and notifications, reducing manual processing time by 70%.
  - Implemented secure authentication and authorization using Spring Security, JWT, and role-based access control (RBAC), ensuring protected access to APIs.

Projects

- **Project A: FinanceApp** *March 2026 - April 2026*
  - Built a full-stack personal finance application using React (Vite) and Spring Boot (Java 17) with RESTful APIs, enabling efficient management of income, expenses, and transactions with scalable CRUD operations.
  - Engineered secure authentication and authorization using Spring Security and JWT, implementing role-based access control (RBAC) to protect endpoints and ensure data integrity.
  - Designed and integrated a financial analytics dashboard with real-time data aggregation, improving user decision-making through clear insights and a responsive UI using Tailwind CSS and animations.
- **Project B: Graphmind** *Feb 2026 - March 2026*
  - Developed an interactive graph visualization platform using React (TypeScript), Vite, and D3-based force-directed graphs, rendering 598+ nodes and 800+ relationships to model SAP Order-to-Cash (O2C) business processes.
  - Integrated an AI-powered natural language query system using Groq (LLaMA 3.3 70B), enabling users to retrieve data insights and dynamically highlight relevant graph nodes from conversational queries.
  - Engineered a responsive, component-based UI with real-time node inspection, interactive legends, and chat interface, improving data exploration and usability for complex enterprise datasets.
- **Project C: NodeCanvas** *Feb 2026 - March 2026*
  - Built a visual pipeline builder using ReactFlow, Zustand, and FastAPI, enabling users to design and manage Directed Acyclic Graphs (DAGs) through an intuitive drag-and-drop interface.
  - Implemented backend DAG validation using Kahn's Algorithm in FastAPI, ensuring correctness of node dependencies and preventing cyclic workflows in real-time.
  - Designed a scalable, modular node system (9+ node types) with a responsive dark-themed UI, improving usability for constructing complex data and AI pipelines.

Skills

- **CS Fundamentals**: OOPs , SQL .
- **Programming Languages**: Java,Python .

Achievements

- Qualified for Level 2: Adobe India Hackathon. *August 2025 - September 2025*
- Qualified for Level 1: Flipkart GRiD 6.0 - Robotics Challenge.
- Qualified for Level 1: Flipkart GRiD 6.0- Software Development Track.

Positions of Responsibility

- **Supervisor,** Literary Club, BITP *Nov 2023 - May 2025*
- **Event Coordinator,** Technika, BITP *Nov 2024 - Dec 2024*